

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 2

Year 12 Higher School Certificate
Trial Examination Term 3 2019

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators and drawing templates may be used
- A reference sheet is provided separately
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination room

Total marks – 100

Section I Pages 3 – 6

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet
provided

Section II Pages 8 – 18

90 marks

Attempt Questions 11 – 16

Start each question in a new writing booklet

Write your student number on every writing booklet

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/15	/15	/15	/15	/15	/15	/100

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

1 i^{2019} simplifies to

(A) 1

(B) -1

(C) i

(D) $-i$

2. The hyperbola $\frac{x^2}{\lambda-3} - \frac{y^2}{\lambda+2} = 1$ has an asymptotic equation $y = \frac{3}{2}x$

The value of λ for this equation is:

(A) -1

(B) 2

(C) 6

(D) 7

3 The locus defined by $z\bar{z} + 3(z + \bar{z}) < 0$ is the region inside the circle

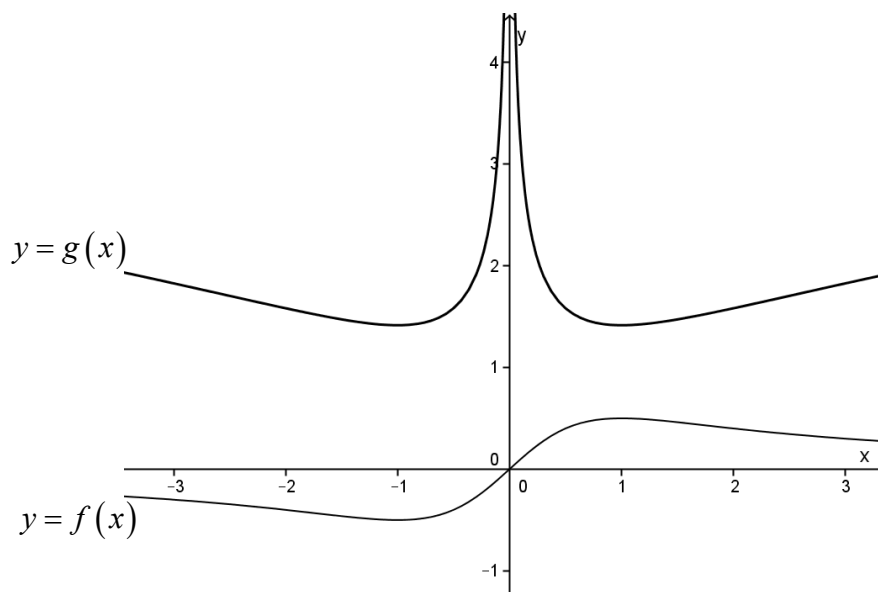
(A) $(x-3)^2 + y^2 = 9$

(B) $(x+3)^2 + y^2 = 9$

(C) $x^2 + (y-3)^2 = 9$

(D) $x^2 + (y+3)^2 = 9$

4. The graphs below are functions $y = f(x)$ and $y = g(x)$ where $y = g(x)$ is the outcome of the original function $y = f(x)$ undergoing a series of transformation.



Select the correct series of transformation involved.

- (A) $g(x) = |x| + \frac{1}{|f(x)|}$
- (B) $g(x) = \frac{1}{|\sqrt{f(x)}|}$
- (C) $g(x) = \frac{1}{\sqrt{f(|x|)}}$
- (D) $g(x) = |x| + \frac{1}{f(|x|)}$
- 5 Find the value of k when $P(x) = x^3 - kx^2 - 10kx + 24$ has a factor of $(x + 2)$.

- (A) 1
- (B) -1
- (C) $\frac{1}{2}$
- (D) $-\frac{1}{2}$

6. The possible roots of $P(x) = a_0x^0 + a_1x^1 + a_2x^2 + \dots + a_{n-1}x^{n-1} + a_nx^n = 0$ could be:

(A) $\pm 1, \pm a_n, \pm \frac{1}{a_n}, \dots$

(B) $\pm 1, \pm a_n, \pm \frac{a_0}{a_n}, \dots$

(C) $\pm 1, \pm a_0, \pm \frac{a_n}{a_0}, \dots$

(D) $\pm 1, \pm a_0, \pm \frac{1}{a_n}, \dots$

7 Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$. Which of the following is correct?

(A) $f(x)$ is even and increasing

(B) $f(x)$ is odd and increasing

(C) $f(x)$ is even and decreasing

(D) $f(x)$ is odd and decreasing

8. If $\int_{-a}^a f(x) dx = 0$ and $\int_0^a f(a-x) dx = \int_0^a f(x) dx$, which of the following functions below possess both of these properties for $a = \pi$?

(A) $f(x) = x \sin^2 x$

(B) $f(x) = x^2 \cos x$

(C) $f(x) = e^x \cos^2 x$

(D) $f(x) = \frac{e^x}{1+e^x} \cos x$

- 9 If a car with mass M , moving with velocity v is opposed by wind resistance αv^2 and road frictional force β , where α and β are constants, then

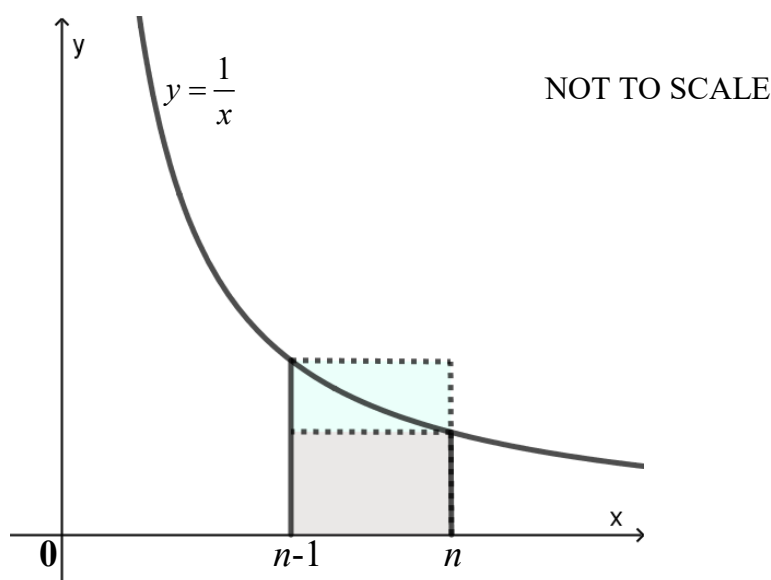
(A) $\frac{dv}{dx} = -\frac{1}{M}(\alpha v + \frac{\beta}{v})$

(B) $\frac{dv}{dx} = \frac{1}{M}(\alpha v + \frac{\beta}{v})$

(C) $\frac{dv}{dx} = -\frac{1}{M}(\alpha v + \beta)$

(D) $\frac{dv}{dx} = \frac{1}{M}(\alpha v + \beta)$

10. Let n be a positive integer greater than 1. Which of the statements below best describe the area of the region under the curve $y = \frac{1}{x}$, $x > 0$ from $x = n-1$ to $x = n$.



(A) $\frac{1}{n} < \ln x < \frac{1}{n+1}$

(B) $\frac{1}{n} \leq \ln x \leq \frac{1}{n+1}$

(C) $e^{-\frac{n}{n-1}} < \left(1 - \frac{1}{n}\right)^n < e^{-1}$

(D) $e^{-\frac{n}{n-1}} \leq \left(1 - \frac{1}{n}\right)^n \leq e^{-1}$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

(a) Let $z = 5 + 3i$ and $w = -3 + 2i$, find in the form $a + ib$ where a and b are real

(i) $\overline{zw}$ 1

(ii) $\frac{2}{iw}$ 2

(b) Find the equations of the asymptotes and vertices of the hyperbola 2

$$\frac{y^2}{12} - \frac{x^2}{4} = 1$$

(c) Sketch the region on the Argand Diagram such that 2

$$2 < |z| < 3 \text{ and } \frac{\pi}{6} < \arg z < \frac{\pi}{2}$$

(d) Use the substitution $t = \tan \frac{x}{2}$ to evaluate 3

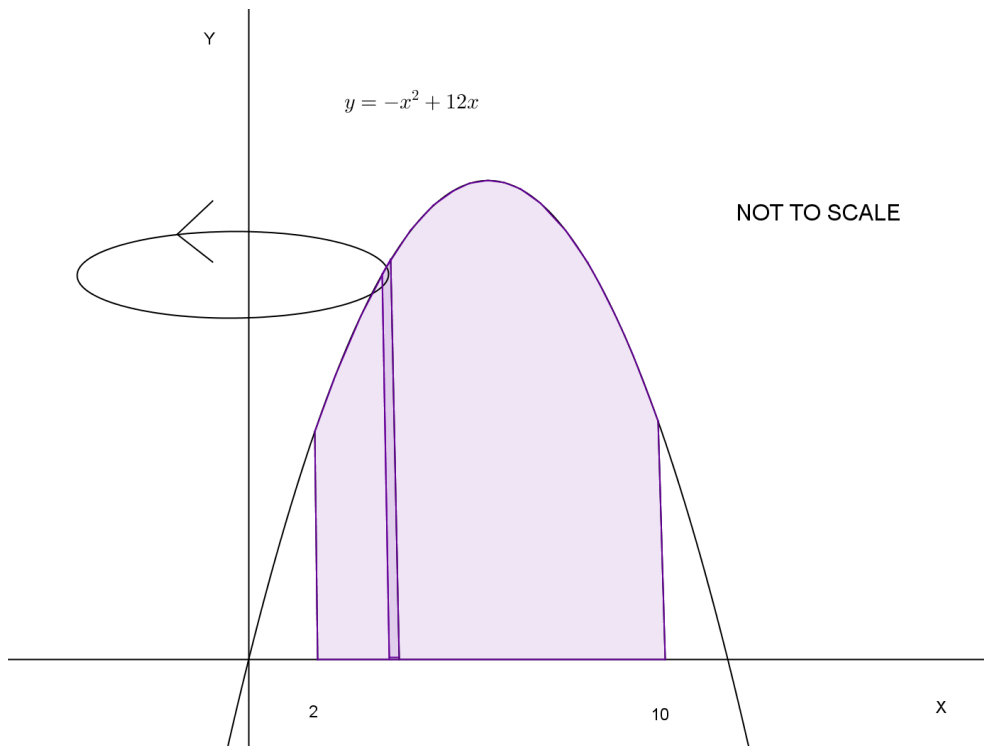
$$\int_0^{\frac{\pi}{2}} \frac{dx}{13 + 5 \sin x + 12 \cos x}$$

Question 11 continues on page 9

Question 11 (continued)

- (e) The region on the diagram below, between the curve $y = 12x - x^2$, the x axis, $x = 2$ and $x = 10$ is rotated about the y axis. Use the method of cylindrical shells to find the volume of the solid.

3



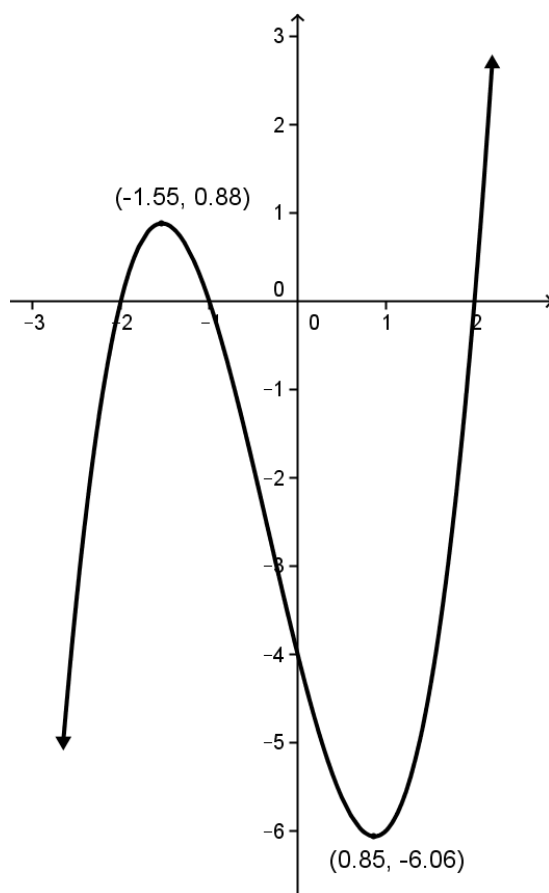
- (f) Evaluate $\int_{-3}^5 \frac{x+7}{\sqrt{x+4}} dx$

2

End of Question 11.

Question 12 (15 marks) Start a new writing booklet

(a) The graph of $y = f(x)$ is shown.



Using the templates provided construct the following transformations of $f(x)$.

(i) $y = f(|x|)$ 1

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = \tan^{-1} f(x)$ 2

(b) (i) Express $\sqrt{8-6i}$ in the form of $a+ib$ where a and b are real and $a > 0$. 2

(ii) Hence solve the quadratic equation $2z^2 + (1-3i)z - 2 = 0$, expressing the answers in the form $c+id$, where c and d are real. 1

Question 12 continues on page 11

Question 12 (continued)

(c) Find $\int_0^{\pi} e^{2x} \sin x \, dx$ **3**

(d) Let $P(x)$ be a polynomial.

(i) Given that $P(x)$ has a root α of multiplicity 3, show that $P(\alpha) = P'(\alpha) = 0$. **2**

(ii) Given that $P(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$ has the factor $(x - 2)^3$. Find the other root. **2**

End of Question 12.

Question 13 (15 marks) Start a new writing booklet

(a) Show that if $x \geq 0, y \geq 0$ then

(i) $x^2 + y^2 \geq 2xy$ 1

(ii) $x^3 + y^3 \geq xy(x + y)$ 2

(iii) hence $2(x^3 + y^3 + z^3) \geq xy(x + y) + yz(y + z) + xz(x + z)$ 2

(b) A particle of Unit mass is projected vertically upwards against gravitational force mg

and resistance $\frac{mv}{k}$ where v is the velocity of the particle and k is a constant.

Thus the motion in the upward direction is given by

$$m\ddot{x} = -mg - \frac{mv}{k}, \quad \text{where } x \text{ is the displacement.}$$

(DO NOT PROVE THIS RESULT)

Initially, the particle has zero displacement and velocity $v_0 = k(h - g)$.

(i) Show that the time (t) of the motion is given by 2

$$t = k \ln\left(\frac{kh}{kg + v}\right)$$

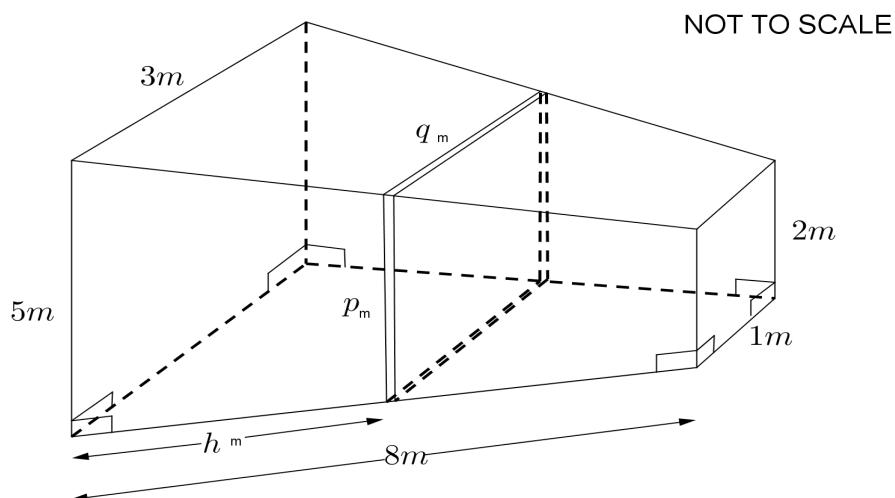
(ii) Show the maximum height (H) of the particle is 3

$$H = k \left[k(h - g) + kg \ln\left(\frac{g}{h}\right) \right]$$

Question 13 continues on page 13

Question 13 (continued)

- (c) A wooden beam of length 8 metres has plane sides with cross-sections parallel to the rectangular ends with dimensions as shown in the diagram below.



- (i) Show $p = 5 - \frac{3h}{8}$ and $q = 3 - \frac{h}{4}$ 2
- (ii) Calculate the area of the cross-section in terms of h 1
- (iii) Calculate the Volume of the beam 2

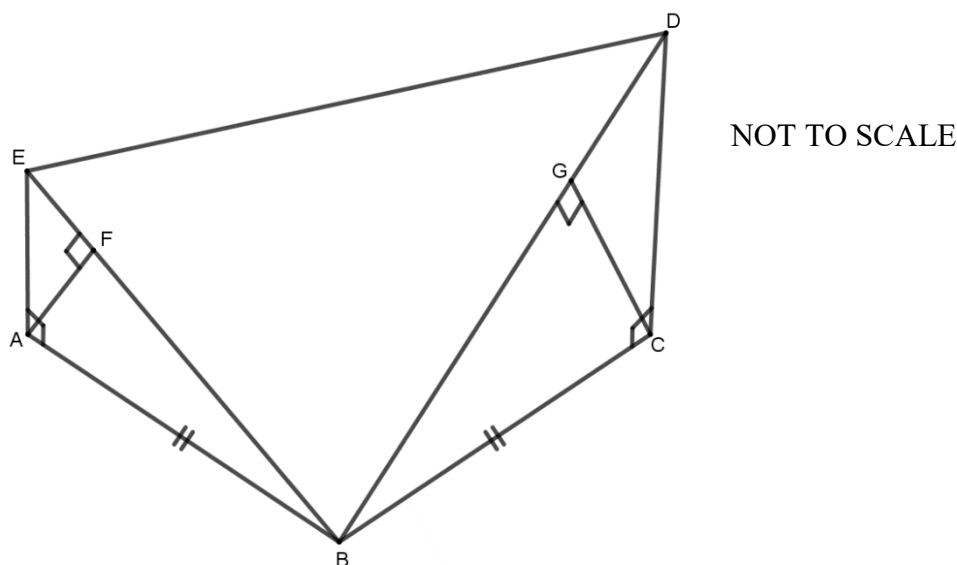
End of Question 13

Question 14 (15 marks)

(a) (i) It is given that $\frac{x}{x^3-8} \equiv \frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4}$. Find the values of A , B and C . 1

(ii) Hence, or otherwise find $\int \frac{x}{x^3-8} dx$. 3

- (b) $ABCDE$ is a two dimensional convex polygon such that $AB = BC$, $\angle BCD = \angle EAB = 90^\circ$, $AF \perp EB$, $BD \perp CG$.



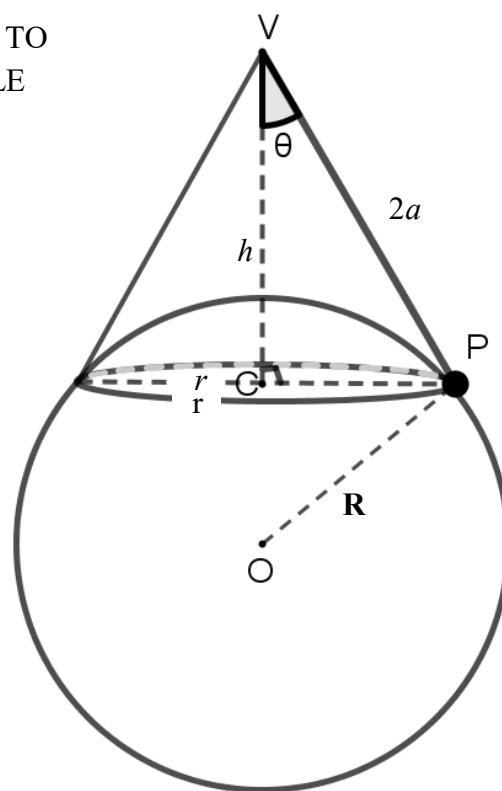
- (i) By using similar triangles prove that $AB^2 = BF \cdot BE$. 1
- (ii) Hence, assuming that $BC^2 = BG \cdot BD$, prove that $\triangle BEG \sim \triangle BDF$. 2
- (iii) Show that $DEFG$ is concyclic. 2

Question 14 (continued)

- (c) A particle P , of mass $2m$ kg, at the end of a light inextensible string of length $2a$ metres, is held h metres at V , vertically above point C , the centre of the circular path of the particle which rests on a smooth sphere of radius R metres.

The string forms a semi vertical angle θ with the vertical. The particle follows a radius r metres on the surface of the sphere with a uniform angular speed of ω radians/second on the outside of the sphere and in contact with it, as shown on the diagram.

NOT TO
SCALE



- (i) Show that the tension (T) in the string, in Newtons is 2
- $$T = 2m(g \cos \theta + 2a\omega^2 \sin^2 \theta).$$
- (ii) Show the normal force (N) on P , in Newtons is 2
- $$N = 2m(g \sin \theta - 2a\omega^2 \cos \theta \sin \theta).$$
- (iii) Show that, for the particle to remain in uniform circular motion on the surface of 2
- the surface of the sphere, then $\omega < \left(\frac{g}{2a \cos \theta} \right)^{\frac{1}{2}}$, where g is acceleration due to gravity.

End of Question 14

Question 15 (15 marks) Start a new writing booklet

(a) Given $I_n = \int_0^1 x^n \sqrt{1-x} dx$ for $n = 1, 2, 3, \dots$

(i) Show that $I_n = \frac{2n}{2n+3} I_{n-1}$ **3**

(ii) Hence Evaluate $\int_0^1 x^3 \sqrt{1-x} dx$ **2**

(b) Consider the curve $x^2 + y^2 + xy = 3$

(i) Show that $\frac{dy}{dx} = -\frac{2x+y}{x+2y}$. **1**

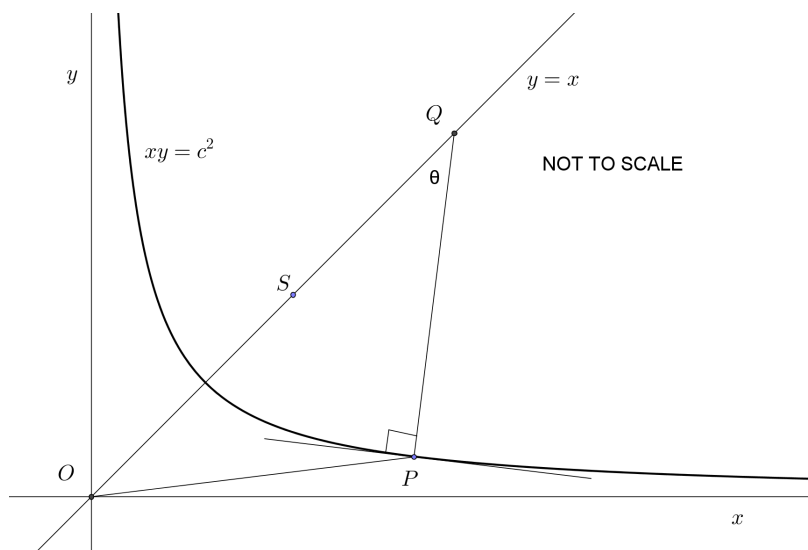
(ii) Deduce the curve has vertical tangents at $(-2,1)$ and $(2,-1)$ and horizontal tangents at $(-1,2)$ and $(1,-2)$. **2**

(iii) Sketch the curve showing these tangents. **2**

Question 15 continues on page 17

Question 15 (continued)

- (c) In the following diagram P is the point $P(ct, \frac{c}{t})$ on the rectangular hyperbola $xy = c^2$, where $t > 0$.



The normal to the hyperbola at P meets the line $y = x$ at Q .

The acute angle between PQ and the line $y = x$ is θ .

S is the focus of the hyperbola nearest to P .

- (i) Show $\tan \theta = \left| \frac{t^2 - 1}{1 + t^2} \right|$. 1
- (ii) Show PQ and PO are equally inclined to $y = x$. 2
- (iii) If PS is perpendicular to $y = x$, show that $\tan \theta = \frac{1}{\sqrt{2}}$ (Hint: consider $\tan^2 \theta$) 2

End of Question 15

Question 16 (15 marks)

- (a) (i) By considering the expansion of $(\cos \theta + i \sin \theta)^5$ and by using De Moivre's Theorem show that

2

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.$$

- (ii) Hence find all the four roots of the equation

2

$$16x^4 - 20x^2 + 5 = 0.$$

- (iii) Hence, or otherwise, show that

3

$$\cos \frac{\pi}{10} \cos \frac{3\pi}{10} = \frac{\sqrt{5}}{4}.$$

- (iv) Find the exact value of

2

$$\sin \frac{3\pi}{5} \sin \frac{6\pi}{5}.$$

- (b) (i) Show that $\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$.

1

- (ii) Hence, or otherwise, find $\int \cos nx \cos mx \, dx$, $n > m > 0$

2

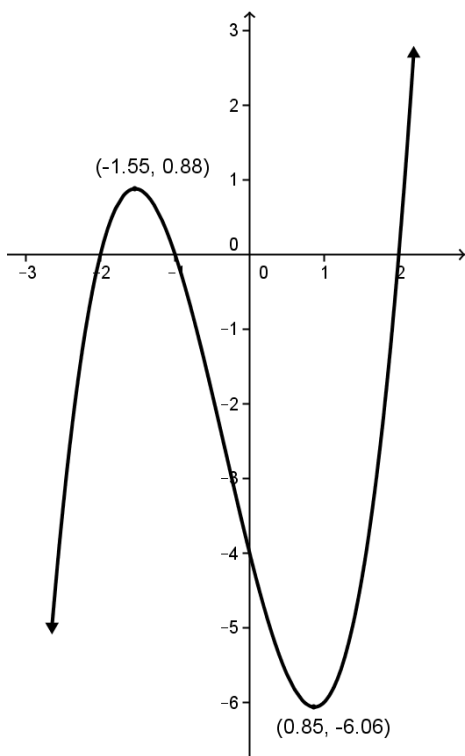
- (iii) Find the exact value of $\sum_{r=1}^9 \int_0^{\frac{\pi}{2}} \sin rx \sin x \, dx$.

3

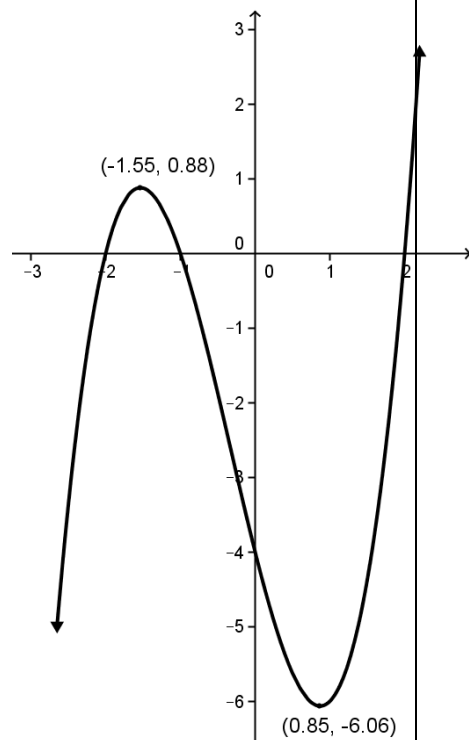
End of Examination

Template for Question 12(a)

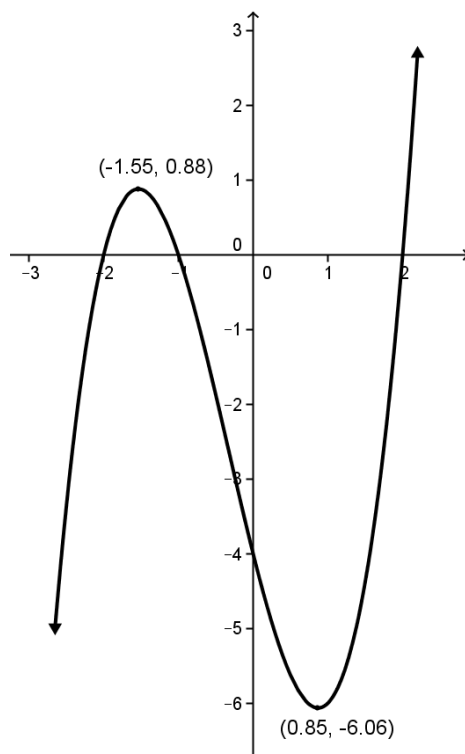
(i) $y = f(|x|)$



(ii) $y = \frac{1}{f(x)}$



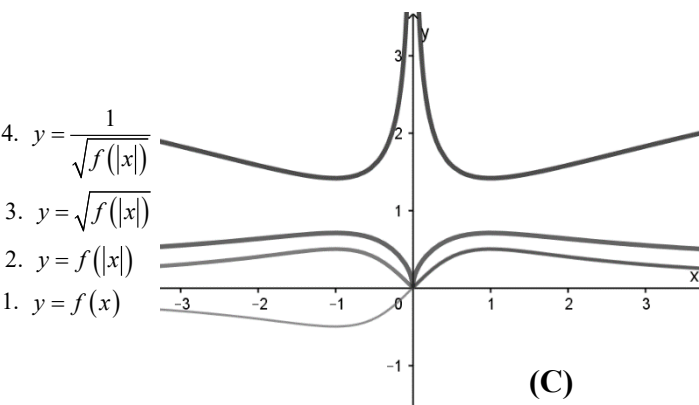
(ii) $y = \tan^{-1} f(x)$



(iii)

Year 12 Mathematics Extension 2 Trial Term 3 2019 Solutions

MULTIPLE CHOICE

Solution	Comment
1. $\frac{2019}{4} = 504 \text{ } r3$ $i^{2019} = i^3$ $= -i$ (D)	
2. $\frac{x^2}{\lambda-3} - \frac{y^2}{\lambda+2} = 1$ $\begin{cases} a^2 = \lambda-3 \\ b^2 = \lambda+2 \end{cases}$ $\frac{b}{a} = \frac{3}{2}$ $\left(\frac{b}{a}\right)^2 = \left(\frac{3}{2}\right)^2$ $\frac{\lambda+2}{\lambda-3} = \frac{9}{4}$ $4(\lambda+2) = 9(\lambda-3)$ $4\lambda+8 = 9\lambda-27$ $35 = 5\lambda$ $\therefore \lambda = 7$ (D)	
3. $z\bar{z} + 3(z + \bar{z}) < 0$ $(x+iy)(x-iy) + 3(x+iy+x-iy) < 0$ $x^2 + y^2 + 6x < 0$ $x^2 + 6x + 9 + y^2 < 9$ $(x+3)^2 + y^2 < 9$ (B)	
4.  (C)	
5. $P(-2) = -8 - 4k + 20k + 24 = 0$ $16k + 16 = 0$ $k = -1$ (B)	

Solution	Comment
6. $P(x) = a_0x^0 + a_1x^1 + a_2x^2 + \dots + a_{n-1}x^{n-1} + a_nx^n = 0$ Possible roots must be of the form $\frac{\pm \text{the factors of } a_0 (i.e. \pm 1, \pm a_0)}{\pm \text{the factors of } a_n (i.e. \pm 1, \pm a_n)} \quad \textbf{(D)}$	
7. $f(x) = \frac{e^x - 1}{e^x + 1}$ $f(-x) = \frac{e^{-x} - 1}{e^{-x} + 1}$ $= \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1}$ $= \frac{1 - e^x}{1 + e^x}$ $= \frac{-e^x}{1 + e^x}$ $= -\frac{e^x}{1 + e^x}$ $f(-x) = -f(x)$ odd $f'(x) = \frac{e^x(e^x + 1) - e^x(e^x - 1)}{(e^x + 1)^2}$ $= \frac{2e^x}{(e^x + 1)^2}$ > 0 for all x (increasing) (B)	
8. $\int_{-a}^a f(x) dx = 0 \rightarrow$ odd function i.e. only A $\int_0^a f(a-x) dx = \int_0^a f(x) dx \rightarrow$ A, B only. (A)	

9.



$$M \ddot{x} = -\alpha v^2 - \beta$$

$$\ddot{x} = -\frac{1}{M}(\alpha v^2 + \beta)$$

$$v \frac{dv}{dx} = -\frac{1}{M}(\alpha v^2 + \beta)$$

$$\frac{dv}{dx} = -\frac{1}{M}\left(\alpha v + \frac{\beta}{v}\right) \quad \text{(A)}$$

10. Area of small rectangle $A_1 = \frac{1}{n} u^2$ Area of large rectangle $A_2 = \frac{1}{n-1} u^2$

$$\int_{n-1}^n \frac{1}{x} dx = [\ln x]_{n-1}^n$$

$$= \ln(n) - \ln(n-1)$$

$$= \ln\left(\frac{n}{n-1}\right)$$

$$A_1 < \int_{n-1}^n \frac{1}{x} dx < A_2$$

$$\frac{1}{n} < \ln \frac{n}{n-1} < \frac{1}{n-1}$$

$$\frac{1}{n} < \ln\left(\frac{n}{n-1}\right) \quad \text{or} \quad \ln\left(\frac{n}{n-1}\right) < \frac{1}{n-1}$$

$$\frac{1}{n} < -\ln \frac{n-1}{n} \quad -\ln\left(\frac{n-1}{n}\right) < \frac{1}{n-1}$$

$$-\frac{1}{n} > \ln\left(1 - \frac{1}{n}\right) \quad \ln\left(1 - \frac{1}{n}\right) > -\frac{1}{n-1}$$

$$e^{-\frac{1}{n}} > \left(1 - \frac{1}{n}\right) \quad \left(1 - \frac{1}{n}\right) > e^{-\frac{1}{n-1}}$$

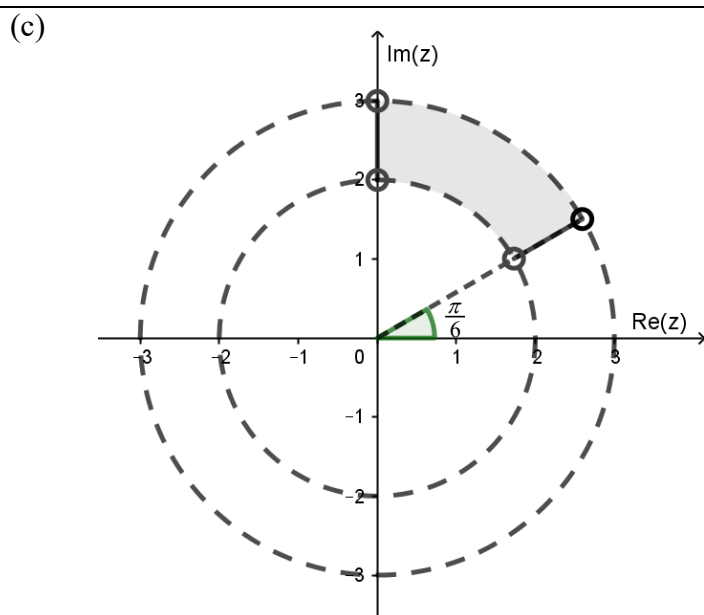
$$\therefore e^{-1} > \left(1 - \frac{1}{n}\right)^n \quad \therefore \left(1 - \frac{1}{n}\right)^n > e^{-\frac{n}{n-1}} \quad \text{(C)}$$

Question 11 Solutions

$$\begin{aligned}
 \text{(a)(i)} \quad z \bar{w} &= (5+3i)(-3-2i) \\
 &= -15-9i-10i-6i^2 \\
 &= -15-9i-10i+6 \\
 &= -9-19i
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{2}{iw} &= \frac{2}{i(-3+2i)} \\
 &= \frac{2}{-3i-2} \times \frac{(-2+3i)}{(-2+3i)} \\
 &= \frac{2(-2+3i)}{4+9} \\
 &= \frac{-4+6i}{13} \\
 &= -\frac{4}{13} + \frac{6i}{13}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \frac{y^2}{12} - \frac{x^2}{4} &= 1 \quad \text{where } a = 2\sqrt{3} \\
 b &= 2 \\
 \text{Equation of asymptotes } x &= \pm \frac{b}{a} y \\
 x &= \pm \frac{2}{2\sqrt{3}} y \\
 x &= \pm \frac{1}{\sqrt{3}} y \\
 \therefore y &= \pm \sqrt{3}x \\
 \therefore \text{Vertices } (0, \pm 2\sqrt{3})
 \end{aligned}$$



Question 11 Solutions

(d) $\int_0^{\frac{\pi}{2}} \frac{dx}{13 + 5 \sin x + 12 \cos x}$ Let $t = \tan \frac{x}{2}$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$
$$dx = \frac{2}{1+t^2} dt$$

When $x = 0$, $t = 0$

$$x = \frac{\pi}{2}, t = 1$$

$$= \int_0^1 \frac{\frac{2}{1+t^2} dt}{13 + 5 \left(\frac{2t}{1+t^2} \right) + 12 \left(\frac{1-t^2}{1+t^2} \right)}$$

$$= \int_0^1 \frac{\frac{2}{1+t^2} dt}{13 + \frac{10t}{1+t^2} + \frac{12-12t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2 dt}{13(1+t^2) + 10t + 12 - 12t^2}$$

$$= \int_0^1 \frac{2 dt}{t^2 + 10t + 25}$$

$$= \int_0^1 \frac{2 dt}{(t+5)^2}$$

$$= \int_0^1 2(t+5)^{-2} dt$$

$$= - \left[2(t+5)^{-1} \right]_0^1$$

$$= \left[\frac{2}{t+5} \right]_1^0$$

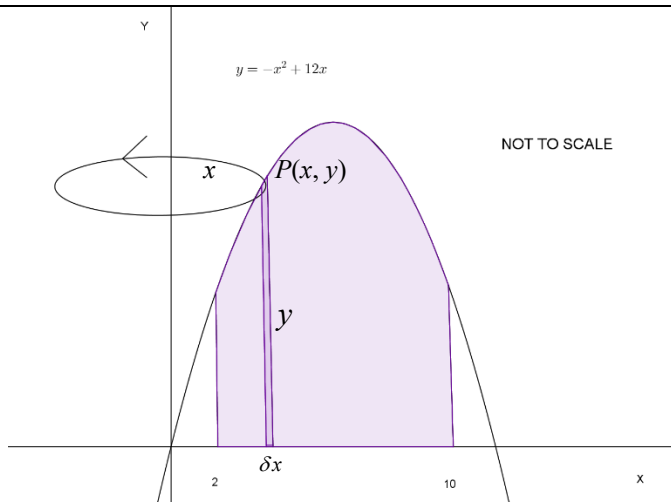
$$= \frac{2}{(0)+5} - \frac{2}{(1)+5}$$

$$= \frac{2}{5} - \frac{1}{3}$$

$$= \frac{1}{15}$$

Question 11 Solutions

(e)



S.A. of hollow cylinder $= 2\pi xy$

$$\delta V = 2\pi xy \delta x \text{ where } y = 12x - x^2$$

$$\delta V = 2\pi x(12x - x^2) \delta x$$

$$\begin{aligned} \text{Volume of solid } V &= \lim_{\delta x \rightarrow 0} \sum_{x=2}^{10} 2\pi x(12x - x^2) \delta x \\ &= 2\pi \int_2^{10} 12x^2 - x^3 \, dx \\ &= 2\pi \left[4x^3 - \frac{x^4}{4} \right]_2^{10} \\ &= 2\pi \left[\left(4000 - \frac{10\,000}{4} \right) - \left(32 - \frac{16}{4} \right) \right] \\ &= 2944\pi \, u^3 \end{aligned}$$

(f)

$$\begin{aligned} &\int_{-3}^5 \frac{x+7}{\sqrt{x+4}} \, dx \\ &= \int_{-3}^5 \frac{x+4+3}{\sqrt{x+4}} \, dx \\ &= \int_{-3}^5 \frac{x+4}{\sqrt{x+4}} + \frac{3}{\sqrt{x+4}} \, dx \\ &= \int_{-3}^5 (x+4)^{\frac{1}{2}} + 3(x+4)^{-\frac{1}{2}} \, dx \\ &= \left[\frac{2(x+4)^{\frac{3}{2}}}{3} + 6(x+4)^{\frac{1}{2}} \right]_{-3}^5 \end{aligned}$$

Question 11 Solutions

(f)cont.

$$\begin{aligned}&= \left[\frac{2(5+4)^{\frac{3}{2}}}{3} + 6(5+4)^{\frac{1}{2}} \right] - \left[\frac{2(-3+4)^{\frac{3}{2}}}{3} + 6(-3+4)^{\frac{1}{2}} \right] \\&= \left[\frac{2(27)}{3} + 6(3) \right] - \left[\frac{2(1)}{3} + 6(1) \right] \\&= 36 - 6\frac{2}{3} \\&= 29\frac{1}{3}\end{aligned}$$

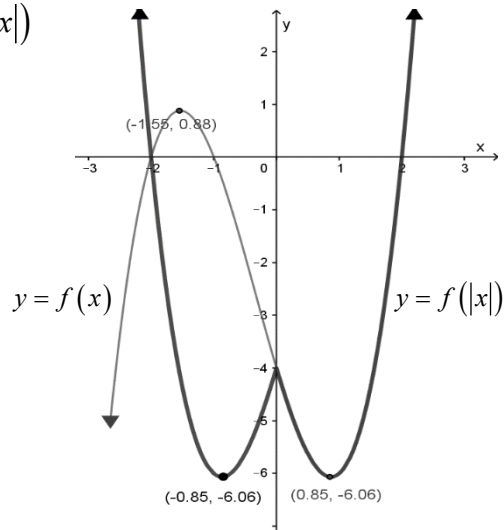
OR $\int_{-3}^5 \frac{x+7}{\sqrt{x+4}} dx$ Let $u^2 = x+4$

$x = u^2 - 4$
 $dx = 2u du$
When $x = 5$, $u = 3$
 $x = -3$, $u = 1$

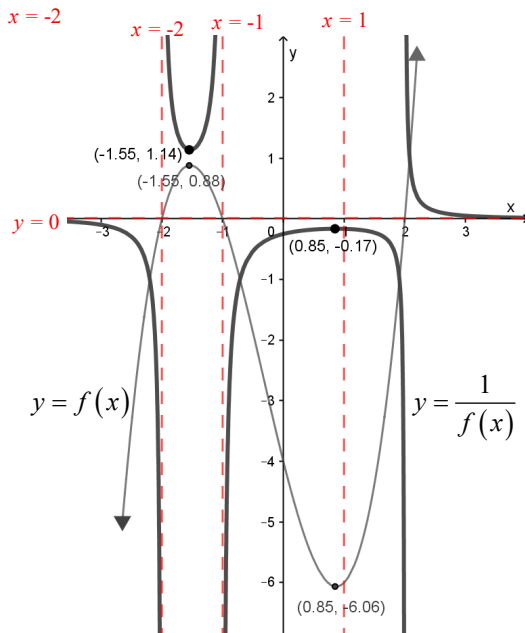
$$\begin{aligned}&= \int_1^3 \frac{u^2+3}{u} 2u du \\&= 2 \int_1^3 (u^2+3) du \\&= 2 \left[\frac{u^3}{3} + 3u \right]_1^3 \\&= 2 \left[\left(\frac{(3)^3}{3} + 3(3) \right) - \left(\frac{(1)^3}{3} + 3(1) \right) \right] \\&= 2 \left[\left(\frac{27}{3} + 9 \right) - \left(\frac{1}{3} + 3 \right) \right] \\&= 2 \left[14\frac{2}{3} \right] \\&= 29\frac{1}{3}\end{aligned}$$

Question 12 Solutions

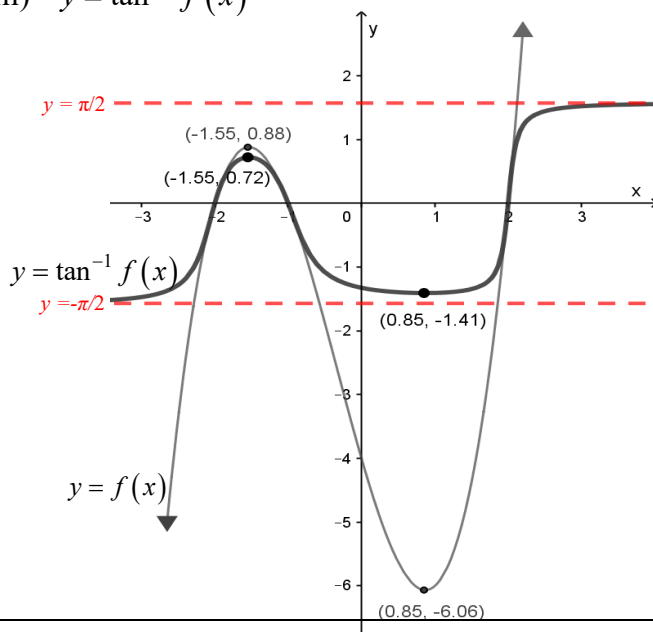
(a)(i) $y = f(|x|)$



(ii) $y = \frac{1}{f(x)}$ $x = -2$



(iii) $y = \tan^{-1} f(x)$



Question 12 Solutions

(b)(i) Let $x + iy = \sqrt{8 - 6i}$

$$(x + iy)^2 = 8 - 6i$$

$$x^2 + 2xyi + y^2i^2 = 8 - 6i$$

$$x^2 - y^2 + 2xyi = 8 - 6i$$

Equating like terms $x^2 - y^2 = 8$ — (1)

$$2xy = -6$$

$$y = -\frac{3}{x} \text{ — (2)}$$

Sub (2) into (1), $x^2 - \left(-\frac{3}{x}\right)^2 = 8$

$$x^2 - \frac{9}{x^2} = 8$$

$$x^4 - 9 = 8x^2$$

$$x^4 - 8x^2 - 9 = 0$$

$$(x^2 - 9)(x^2 + 1) = 0$$

Since x is the real component of the complex number,

$$\therefore x^2 = 9$$

$$\therefore x = \pm 3$$

Sub $x = \pm 3$ into (2), $\therefore y = \mp 1$

$$\therefore \sqrt{8 - 6i} = 3 - i \text{ or } -3 + i$$

(ii) $2z^2 + (1 - 3i)z - 2 = 0$

$$a = 2 \quad b = 1 - 3i \quad c = -2$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(1 - 3i) \pm \sqrt{(1 - 3i)^2 - 4(2)(-2)}}{2(2)}$$

$$= \frac{-1 + 3i \pm \sqrt{1 - 6i - 9 + 16}}{4}$$

$$= \frac{-1 + 3i \pm \sqrt{8 - 6i}}{4}$$

$$z = \frac{-1 + 3i + (3 - i)}{4} \text{ or } = \frac{-1 + 3i + (-3 + i)}{4}$$

$$z = \frac{2 + 2i}{4} \text{ or } = \frac{-4 + 4i}{4}$$

$$\therefore z = \frac{1}{2} + \frac{1}{2}i \text{ or } -1 + i$$

Question 12 Solutions

(c) Let $I = \int_0^{\pi} e^{2x} \sin x \, dx$

$$= [uv]_0^{\pi} - \int_0^{\pi} u'v \, dx$$

where $u = e^{2x}$ and $v' = \sin x$
 $u' = 2e^{2x}$ $v = -\cos x$

$$= [-e^{2x} \cos x]_0^{\pi} - (-)2 \int_0^{\pi} e^{2x} \cos x \, dx$$

$$= [-e^{2(\pi)} \cos(\pi) - (-e^{2(0)} \cos(0))] + 2 \int_0^{\pi} e^{2x} \cos x \, dx$$

$$= [-e^{2\pi}(-1) + (1 \times 1)] + 2 \int_0^{\pi} e^{2x} \cos x \, dx$$

$$= (e^{2\pi} + 1) + 2 \int_0^{\pi} e^{2x} \cos x \, dx$$

$$= (e^{2\pi} + 1) + 2 \left[[uv]_0^{\pi} - \int_0^{\pi} u'v \, dx \right]$$

where $u = e^{2x}$ and $v' = \cos x$
 $u' = 2e^{2x}$ $v = \sin x$

$$= (e^{2\pi} + 1) + 2 \left[[e^{2x} \sin x]_0^{\pi} - 2 \int_0^{\pi} e^{2x} \sin x \, dx \right]$$

$$= (e^{2\pi} + 1) + 2 \left[[e^{2(\pi)} \sin(\pi) - e^{2(0)} \sin(0)] - 2I \right]$$

$$= (e^{2\pi} + 1) + 2 \left[[0 - 0] - 2I \right]$$

$$I = (e^{2\pi} + 1) - 4I$$

$$5I = (e^{2\pi} + 1)$$

$$\therefore I = \frac{e^{2\pi} + 1}{5}$$

(d) Let $P(x) = (x - \alpha)^3 Q(x)$ with α being the root of multiplicity of 3

$$P(\alpha) = (\alpha - \alpha)^3 Q(\alpha)$$

$$P(\alpha) = (0)^3 Q(\alpha)$$

$$\therefore P(\alpha) = 0$$

Question 12 Solutions

(d)(i)cont. $P(x) = (x - \alpha)^3 Q(x)$

$$P'(x) = u'v + v'u \quad \text{where } u = (x - \alpha)^3$$

$$u' = 3(x - \alpha)^2$$

$$v = Q(x)$$

$$v' = Q'(x)$$

$$P'(x) = 3(x - \alpha)^2 Q(x) + (x - \alpha)^3 Q'(x)$$

$$= (x - \alpha)^2 [3Q(x) + (x - \alpha)Q'(x)]$$

$$\text{and let } M(x) = 3Q(x) + (x - \alpha)Q'(x)$$

$$\therefore P'(x) = (x - \alpha)^2 M(x)$$

$$P''(x) = u'v + v'u \quad \text{where } u = (x - \alpha)^2$$

$$u' = 2(x - \alpha)$$

$$v = M(x)$$

$$v' = M'(x)$$

$$P''(x) = 2(x - \alpha)M(x) + (x - \alpha)^2 M'(x)$$

$$= (x - \alpha)[2M(x) + (x - \alpha)M'(x)]$$

$$\therefore P''(\alpha) = (\alpha - \alpha)[2M(\alpha) + (\alpha - \alpha)M'(\alpha)]$$

$$\therefore P''(\alpha) = (0)[2M(\alpha) + (0)M'(\alpha)]$$

$$\therefore P''(\alpha) = 0$$

$$\therefore P(\alpha) = P''(\alpha) = 0 \quad (\text{as required})$$

(ii) $P(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$ divided by factor $(x - 2)^3$.

Method 1: $(x - 2)^3 = x^3 - 3x^2(2) + 3x(2)^2 - 8$

$$= x^3 - 6x^2 + 12x - 8$$

$$x + 1$$

$$x^3 - 6x^2 + 12x - 8 \overline{) x^4 - 5x^3 + 6x^2 + 4x - 8}$$

$$- (x^4 - 6x^3 + 12x^2 - 8x) \downarrow$$

$$x^3 - 6x^2 + 12x - 8$$

$$- (x^3 - 6x^2 + 12x - 8)$$

$$0$$

$\therefore$ the remainder root is -1.

Method 2: Let the roots be 2, 2, 2, α

$$2(2)(2)\alpha = \frac{e}{a}$$

Question 12 Solutions	
(d)(ii)cont. $8\alpha = \frac{(-8)}{(1)}$ $8\alpha = -8$ $\therefore \alpha = -1 \text{ i.e. the remainder root}$ $8\alpha = -8$ $\therefore \alpha = -1 \text{ i.e. the remainder root}$	

Question 13 Solutions	
(a)(i) $(x - y)^2 \geq 0$ $x^2 - 2xy + y^2 \geq 0$ $x^2 + y^2 \geq 2xy \quad (\text{as required})$ (ii) $x^2 + y^2 \geq 2xy$ $(x^2 + y^2)(x + y) \geq 2xy(x + y).$ $x^3 + xy^2 + x^2y + y^3 \geq 2xy(x + y)$ $x^3 + y^3 + xy(x + y) \geq 2xy(x + y)$ $\therefore x^3 + y^3 \geq xy(x + y) \quad (\text{as required})$ (iii) Since $x^3 + y^3 \geq xy(x + y)$ Similarly $x^3 + z^3 \geq xz(x + z)$ $z^3 + y^3 \geq zy(z + y)$ $\therefore 2(x^3 + y^3 + z^3) \geq xy(x + y) + xz(x + z) + zy(z + y)$ (as required)	
(b)(i) $m \ddot{x} = -mg - \frac{mv}{k}$ $m \ddot{x} = -m \left(g + \frac{v}{k} \right)$ $\ddot{x} = - \left(g + \frac{v}{k} \right)$ $\frac{dv}{dt} = - \left(g + \frac{v}{k} \right)$ $\frac{dv}{dt} = - \left(\frac{gk + v}{k} \right)$ $\frac{dt}{dv} = - \left(\frac{k}{gk + v} \right)$ $- dt = \frac{k dv}{gk + v}$	

Question 13 Solutions

$$(b)(i) \text{cont.} \quad \int_0^t dt = - \int_{v_0}^v \frac{k}{gk + v} dv$$

$$t = -k \left[\ln(gk + v) \right]_{v_0}^v$$

$$= -k \left[\ln(gk + v) - \ln(gk + v_0) \right]$$

$$= -k \left[\ln \left(\frac{gk + v}{gk + v_0} \right) \right]$$

$$= k \ln \left(\frac{gk + v_0}{gk + v} \right) \text{ since } v_0 = k(h - g)$$

$$\therefore t = k \ln \left(\frac{kh}{gk + v} \right) \quad (\text{as required})$$

(ii) Time to reach max height is when $v = 0$,

$$t = k \ln \left(\frac{gh}{gk + (0)} \right).$$

$$\therefore t = k \ln \left(\frac{h}{g} \right)$$

Max height $\ddot{x} = - \left(g + \frac{v}{k} \right)$

$$v \frac{dv}{dx} = - \left(\frac{gk + v}{k} \right)$$

$$\frac{dv}{dx} = - \left(\frac{gk + v}{kv} \right)$$

$$\frac{dx}{dv} = - \left(\frac{kv}{gk + v} \right)$$

$$\frac{dx}{k} = - \left(\frac{v}{gk + v} \right) dv$$

$$\int_0^H \frac{dx}{k} = - \int_{v_0}^0 \left(\frac{v}{gk + v} \right) dv$$

$$\left[\frac{x}{k} \right]_0^H = \int_0^{v_0} \left(\frac{gk + v - gk}{gk + v} \right) dv$$

$$\frac{H}{k} = \int_0^{v_0} \left(1 - \frac{gk}{gk + v} \right) dv$$

$$\frac{H}{k} = \left[v - gk \ln(gk + v) \right]_0^{v_0}$$

$$H = k \left[v_0 - gk \ln(gk + v_0) - (-gk \ln(gk)) \right]$$

$$H = k \left[v_0 - gk \ln \left(\frac{gk + v_0}{gk} \right) \right]$$

Question 13 Solutions

(b)(ii)cont. $H = k \left[v_0 - gk \ln(gk + v_0) + gk \ln(gk) \right]$

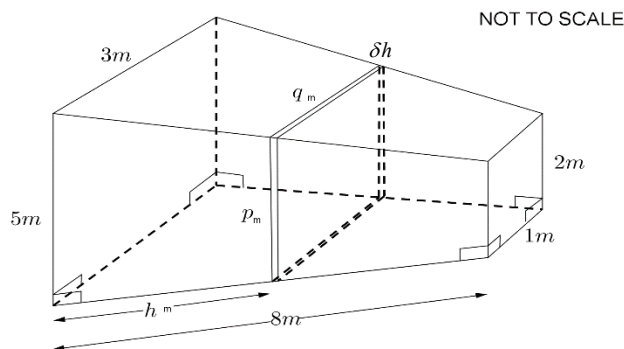
$$H = k \left[v_0 + gk \ln \left(\frac{gk}{gk + v_0} \right) \right]$$

Since $v_0 = kh - kg$

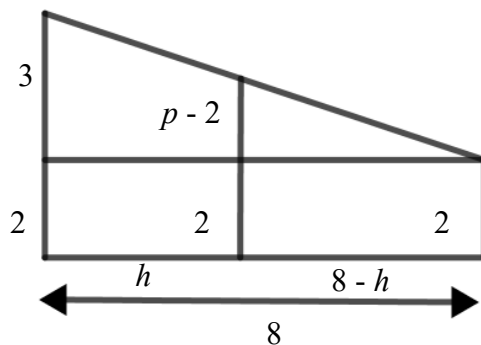
$$= k \left[(kh - kg) + gk \ln \left(\frac{gk}{gk + kh - kg} \right) \right]$$

$$\therefore H = k \left[(kh - kg) + gk \ln \left(\frac{g}{h} \right) \right] \quad (\text{as required})$$

(c)



(i)

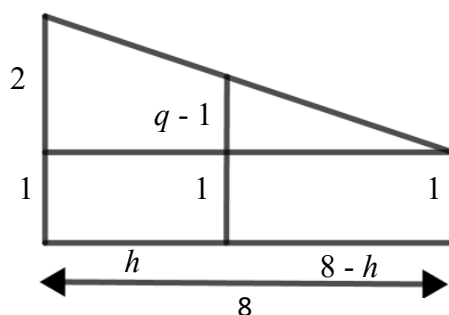


$$\frac{p-2}{3} = \frac{8-h}{8}$$

$$8p - 16 = 24 - 3h$$

$$8p = 40 - 3h$$

$$\therefore p = 5 - \frac{3h}{8} \quad (\text{as required})$$



Question 13 Solutions

(c)(i)cont. $\frac{q-1}{2} = \frac{8-h}{8}$
 $8q-8=16-2h$
 $8q=24-2h$
 $\therefore q=3-\frac{h}{4}$ (as required)

(ii) $A_{pq} = pq$
 $= \left(5 - \frac{3h}{8}\right) \left(3 - \frac{h}{4}\right)$
 $= 15 - \frac{5h}{4} - \frac{9h}{8} + \frac{3h^2}{32}$
 $= 15 - \frac{19h}{8} + \frac{3h^2}{32}$

(iii) $\delta V = A_{pq} \delta h$
 $V = \lim_{\delta h \rightarrow 0} \sum_{h=0}^8 \left(15 - \frac{19h}{8} + \frac{3h^2}{32}\right) \delta h$
 $= \int_0^8 \left(15 - \frac{19h}{8} + \frac{3h^2}{32}\right) dh$
 $= \left[15h - \frac{19h^2}{16} + \frac{h^3}{32}\right]_0^8$
 $= \left[15(8) - \frac{19(8)^2}{16} + \frac{(8)^3}{32}\right] - 0$
 $= 120 - 76 + 16$
 $= 60 \text{ m}^3$

Question 14 Solutions

(a)(i) $\frac{x}{x^3-8} \equiv \frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4}$
 $x \equiv A(x^2+2x+4) + (Bx+C)(x-2)$
 $x \equiv Ax^2 + 2Ax + 4A + (Bx^2 - 2Bx + Cx - 2C)$
 $x \equiv Ax^2 + 2Ax + 4A + Bx^2 - 2Bx + Cx - 2C$
 $ + Cx$
 $x \equiv (A+B)x^2 + (2A+C-2B)x + (4A-2C)$
 Equating like terms: $A+B=0$
 $\phantom{\text{Equating like terms:}} B=-A \quad \text{--- (1)}$
 $2A+C-2B=1 \quad \text{--- (2)}$

Question 14 Solutions

(a)(ii)cont.

$$4A - 2C = 0$$

$$2C = 4A$$

$$C = 2A \quad \text{--- (3)}$$

$$\text{Sub (1) and (3) into (2)} \quad 2A + (2A) - 2(-A) = 1$$

$$6A = 1$$

$$A = \frac{1}{6} \quad (4)$$

$$\text{Sub (4) into (1)} \quad \therefore B = -\frac{1}{6}$$

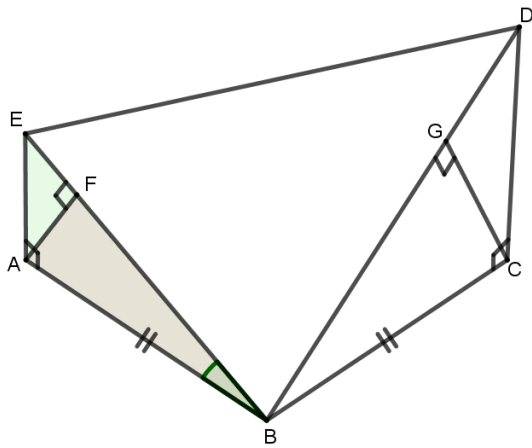
$$\text{Sub (4) into (3)} \quad \therefore C = \frac{2}{6}$$

$$\therefore A = \frac{1}{6}, B = -\frac{1}{6} \text{ and } C = \frac{1}{3}$$

$$\begin{aligned}
 \text{(ii)} \quad & \int \frac{x}{x^3 - 8} dx \\
 &= \frac{1}{6} \int \left(\frac{1}{x-2} + \frac{-x+2}{x^2+2x+4} \right) dx \\
 &= \frac{1}{6} \int \left(\frac{1}{x-2} - \frac{x-2}{x^2+2x+4} \right) dx \\
 &= \frac{1}{6} \int \left(\frac{1}{x-2} - \frac{(x+1)-2-1}{x^2+2x+4} \right) dx \\
 &= \frac{1}{6} \int \left(\frac{1}{x-2} - \frac{(x+1)-3}{x^2+2x+4} \right) dx \\
 &= \frac{1}{6} \left[\int \left(\frac{1}{x-2} - \frac{(x+1)}{x^2+2x+4} + \frac{3}{x^2+2x+4} \right) dx \right] \\
 &= \frac{1}{6} \left[\int \left(\frac{1}{x-2} - \frac{1}{2} \cdot \frac{2(x+1)}{x^2+2x+4} + \frac{3}{(x^2+2x+1)+3} \right) dx \right] \\
 &= \frac{1}{6} \left[\int \left(\frac{1}{x-2} - \frac{1}{2} \cdot \frac{2x+2}{x^2+2x+4} + \frac{3}{(x+1)^2+3} \right) dx \right] \\
 &= \frac{1}{6} \left[\ln|x-2| - \frac{1}{2} \ln(x^2+2x+4) + \frac{3}{\sqrt{3}} \int \left(\frac{\sqrt{3}}{(x+1)^2+3} \right) dx \right] \\
 &= \frac{1}{6} \left[\ln|x-2| - \frac{1}{2} \ln(x^2+2x+4) + \sqrt{3} \int \left(\frac{\sqrt{3}}{(x+1)^2+3} \right) dx \right] \\
 &= \frac{\ln|x-2|}{6} - \frac{\ln(x^2+2x+4)}{12} + \frac{\sqrt{3}}{6} \tan^{-1} \frac{x+1}{\sqrt{3}} + C
 \end{aligned}$$

Question 14 Solutions

(b)(i)



In $\triangle BAE$ and $\triangle BFA$,

$$\angle BAE = \angle BFA = 90^\circ$$

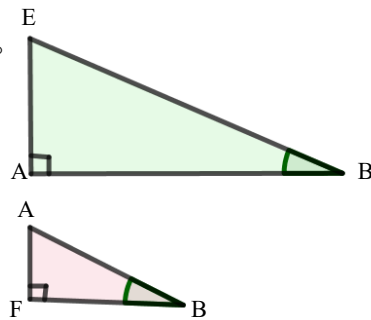
(given)

$$\angle ABE = \angle FBA$$

(common)

$$\therefore \triangle BAE \sim \triangle BFA$$

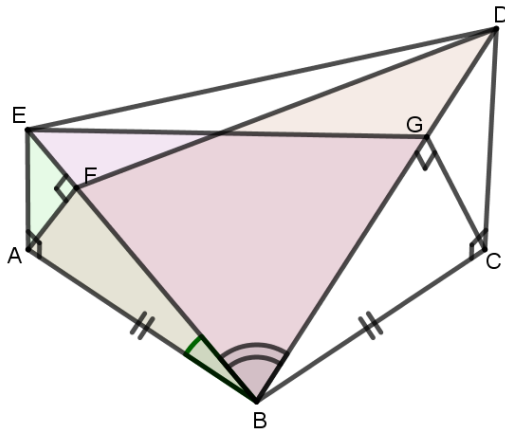
(equiangular)



$$\frac{BE}{BA} = \frac{BA}{BF} \quad (\text{corresponding sides of similar triangles in same ratio})$$

$$\therefore BE \cdot BF = BA^2$$

(ii) Assume $BC^2 = BG \cdot BD$,



$$AB = CB \quad (\text{given})$$

$$AB^2 = CB^2$$

$$BF \cdot BE = BG \cdot BD \quad [\text{from (a) and given}]$$

$$\frac{BG}{BF} = \frac{BE}{BD}$$

In $\triangle BEG$ and $\triangle BDF$,

$$\angle EBG = \angle DBF \quad (\text{common})$$

$$\frac{BG}{BF} = \frac{BE}{BD} \quad (\text{proven above})$$

Question 14 Solutions

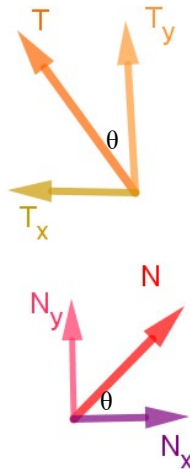
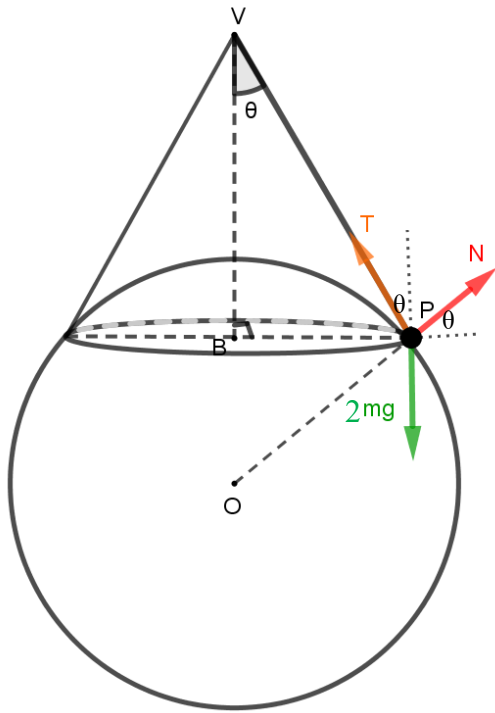
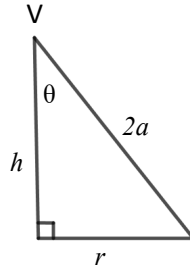
(b)(ii)cont. $\therefore \triangle BEG \parallel \triangle BDF$ (two sides in same ratio and an equal included angle)

(iii) In $\triangle BEG$ and $\triangle BDF$,
 $\angle BEG = \angle BDF$ (corresponding angles of similar triangles)
 $\therefore D, E, F$ and G are concyclic points (angles in the same segment)
 $\therefore DEFG$ is a cyclic quadrilateral.

(c)(i) In $\triangle VBP$,

$$r = 2a \sin \theta$$

$$h = 2a \cos \theta$$



Vertically: $T \cos \theta + N \sin \theta = 2mg$ (1)

Horizontally: $T \sin \theta - N \cos \theta = 2mr\omega^2$ (2)

(1) $\times \cos \theta$ $T \cos^2 \theta + N \sin \theta \cos \theta = 2mg \cos \theta$ (1)'

(2) $\times \sin \theta$ $T \sin^2 \theta - N \sin \theta \cos \theta = 2mr\omega^2 \sin \theta$ (2)'

(1)' + (2)' $T \cos^2 \theta + T \sin^2 \theta = 2mg \cos \theta + 2mr\omega^2 \sin \theta$

$$T(\cos^2 \theta + \sin^2 \theta) = 2m(g \cos \theta + r\omega^2 \sin \theta)$$

$$\therefore T = 2m(g \cos \theta + r\omega^2 \sin \theta)$$

$$= 2m(g \cos \theta + (2a \sin \theta)\omega^2 \sin \theta)$$

$$\therefore T = 2m(g \cos \theta + 2a\omega^2 \sin^2 \theta) \text{ Newtons}$$

Question 14 Solutions

(c)(ii) (1) $\times \sin \theta \quad T \cos \theta \sin \theta + N \sin^2 \theta = 2mg \sin \theta \quad (1)''$
 (2) $\times \cos \theta \quad T \sin \theta \cos \theta - N \cos^2 \theta = 2mr\omega^2 \cos \theta \quad (2)''$
 (1)''-(2)'' $N \cos^2 \theta + N \sin^2 \theta = 2mg \sin \theta - 2mr\omega^2 \cos \theta$
 $N(\cos^2 \theta + \sin^2 \theta) = 2m(g \sin \theta - r\omega^2 \cos \theta)$
 $\therefore N = 2m(g \sin \theta - r\omega^2 \cos \theta)$
 $\therefore N = 2m(g \sin \theta - 2a\omega^2 \cos \theta \sin \theta) \text{ Newton}$

(iii) For the particle to remain in contact with the surface of the sphere, then $T > 0$ and $N > 0$ for all the values of ω . Since T is always positive, thus need to consider N .

Hence $N > 0$
 $g \sin \theta - 2a\omega^2 \sin \theta \cos \theta > 0$
 $\sin \theta (g - 2a\omega^2 \cos \theta) > 0$ and since $0^\circ < \theta < 90^\circ$
 $\therefore \sin \theta > 0$
 i.e. $g - 2a\omega^2 \cos \theta > 0$
 $g > 2a\omega^2 \cos \theta$
 $\frac{g}{2a \cos \theta} > \omega^2$
 $\therefore \omega^2 < \frac{g}{2a \cos \theta}$
 $\therefore \omega < \left(\frac{g}{2a \cos \theta} \right)^{\frac{1}{2}}$

Question 15 Solutions

(a)(i) $I_n = \int_0^1 x^n \sqrt{1-x} \, dx$ for $n = 0, 1, 2, \dots$
 i.e. $I_{n-1} = \int_0^1 x^{n-1} \sqrt{1-x} \, dx$
 $= [uv]_0^1 - \int_0^1 u'v \, dx$ where $u = x^n$ and $v' = \sqrt{1-x}$
 $u' = nx^{n-1} \quad v = \frac{2(1-x)^{\frac{3}{2}}}{-3}$
 $= \left[-\frac{2(1-x)^{\frac{3}{2}} x^n}{3} \right]_0^1 - n \int_0^1 x^{n-1} \left(-\frac{2(1-x)^{\frac{3}{2}}}{3} \right) dx$

Question 15 Solutions

$$\begin{aligned} \text{(a)(i)cont.} \quad &= \left[\frac{2(1-x)^{\frac{3}{2}} x^n}{3} \right]_1^0 + \frac{2n}{3} \int_0^1 x^{n-1} (1-x) \sqrt{1-x} \, dx \\ &= \left[\frac{2(1-0)^{\frac{3}{2}} (0)^n}{3} - \frac{2(1-1)^{\frac{3}{2}} (1)^n}{3} \right] + \frac{2n}{3} \int_0^1 (x^{n-1} - x^n) \sqrt{1-x} \, dx \end{aligned}$$

$$= 0 + \frac{2n}{3} \int_0^1 (x^{n-1} \sqrt{1-x} - x^n \sqrt{1-x}) \, dx$$

$$= \frac{2n}{3} \int_0^1 x^{n-1} \sqrt{1-x} \, dx - \frac{2n}{3} \int_0^1 x^n \sqrt{1-x} \, dx$$

$$I_n = \frac{2n}{3} I_{n-1} - \frac{2n}{3} I_n$$

$$I_n + \frac{2n}{3} I_n = \frac{2n}{3} I_{n-1}$$

$$\frac{3+2n}{3} I_n = \frac{2n}{3} I_{n-1}$$

$$I_n = \frac{2n}{3} \left(\frac{3}{3+2n} \right) I_{n-1}$$

$$\therefore I_n = \frac{2n}{3+2n} I_{n-1} \quad (\text{as required})$$

$$\text{(ii) Let } I_3 = \int_0^1 x^3 \sqrt{1-x} \, dx$$

$$= \frac{2(3)}{3+2(3)} I_2$$

$$= \frac{6}{9} I_2$$

$$= \frac{2}{3} I_2$$

$$I_2 = \frac{2(2)}{3+2(2)} I_1$$

$$= \frac{4}{7} I_1$$

$$I_1 = \frac{2(1)}{3+2(1)} I_0$$

$$= \frac{2}{5} I_0$$

$$I_0 = \int_0^1 x^0 \sqrt{1-x} \, dx$$

Question 15 Solutions

(a)(ii)cont. $I_0 = \int_0^1 (1-x)^{\frac{1}{2}} dx$

$$= \left[-\frac{2(1-x)^{\frac{3}{2}}}{3} \right]_0^1$$
$$= \frac{2}{3} \left[(1-x)^{\frac{3}{2}} \right]_1^0$$
$$= \frac{2}{3} \left[(1-0)^{\frac{3}{2}} - (1-1)^{\frac{3}{2}} \right]$$
$$= \frac{2}{3} [1-0]$$
$$\therefore I_0 = \frac{2}{3}$$
$$\therefore I_1 = \frac{2}{5} \left(\frac{2}{3} \right)$$
$$= \frac{4}{15}$$
$$\therefore I_2 = \frac{4}{7} \left(\frac{4}{15} \right)$$
$$= \frac{16}{105}$$
$$\therefore I_3 = \frac{2}{3} \left(\frac{16}{105} \right)$$
$$\therefore I_3 = \frac{32}{315}$$

(b)(i) $x^2 + y^2 + xy = 3$

$$2x + 2y \frac{dy}{dx} + u'v + v'u = 0 \text{ where } u = x \text{ and } v = y$$
$$u' = 1 \quad v' = \frac{dy}{dx}$$
$$2x + 2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$$
$$2x + y + 2y \frac{dy}{dx} + x \frac{dy}{dx} = 0$$
$$2x + y + (2y + x) \frac{dy}{dx} = 0$$
$$(2y + x) \frac{dy}{dx} = -(2x + y)$$
$$\frac{dy}{dx} = -\frac{(2x + y)}{x + 2y} \quad (\text{as required})$$

Question 15 Solutions

(b)(ii) For vertical tangents when $x + 2y = 0$
 $x = -2y$

$$(-2y)^2 + y^2 + (-2y)y = 3$$

$$4y^2 + y^2 - 2y^2 = 3$$

$$3y^2 = 3$$

$$y^2 = 1$$

$$\therefore y = \pm 1 \Rightarrow x = \mp 2$$

$\therefore$ Vertical tangents at $(2, -1)$ and $(-2, 1)$.

For horizontal tangents when $2x + y = 0$

$$y = -2x$$

$$x^2 + (-2x)^2 + x(-2x) = 3$$

$$x^2 + 4x^2 - 2x^2 = 3$$

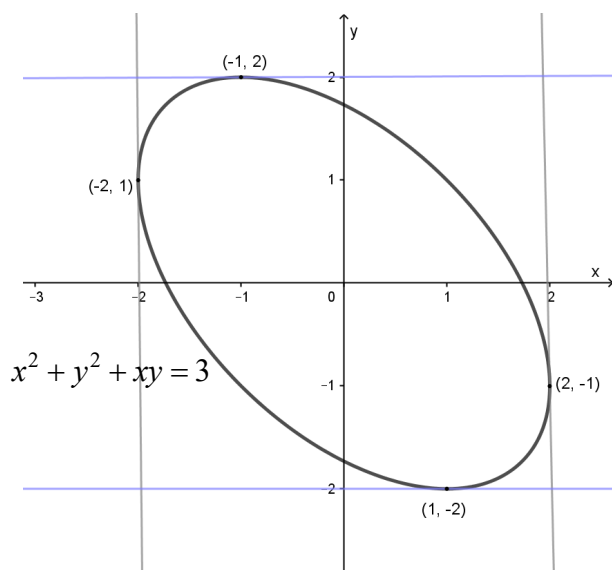
$$3x^2 = 3$$

$$x^2 = 1$$

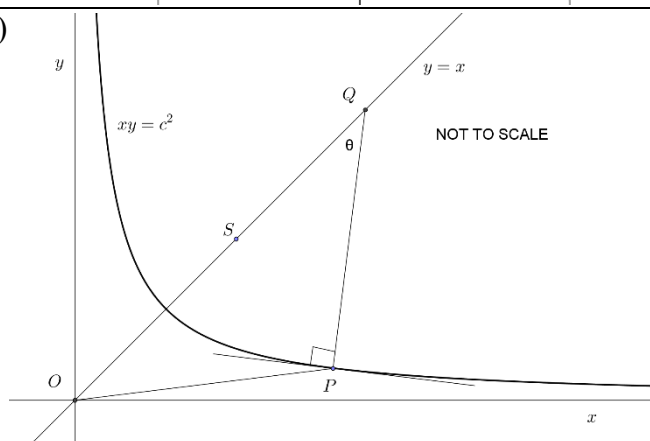
$$\therefore x = \pm 1 \Rightarrow y = \mp 2$$

$\therefore$ Horizontal tangents at $(1, -2)$ and $(-1, 2)$.

(iii)



(c)(i)



Question 15 Solutions

(c)(i)cont. $x = ct$ and $y = \frac{c}{t}$

$$\frac{dx}{dt} = c \quad \frac{dy}{dt} = -\frac{c}{t^2}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} \\ &= -\frac{c}{t^2} \times \frac{1}{c} \\ &= -\frac{1}{t^2}\end{aligned}$$

Since tangent at $P \perp PQ$, $\therefore m_{PQ} = t^2$

$$\tan \theta = \left| \frac{m_{PQ} - m_{OQ}}{1 + m_{PQ} \times m_{OQ}} \right|$$

$$\therefore \tan \theta = \left| \frac{t^2 - 1}{1 + t^2} \right| \quad (\text{as required})$$

(ii) $m_{PO} = \frac{\frac{c}{t} - 0}{ct - 0}$

$$\begin{aligned}&= \frac{\left(\frac{c}{t}\right)}{ct} \\ &= \frac{1}{t^2}\end{aligned}$$

$$\begin{aligned}\tan \angle POQ &= \left| \frac{1 - \frac{1}{t^2}}{1 + \frac{1}{t^2}} \right| \\ &= \left| \frac{\frac{t^2 - 1}{t^2}}{\frac{t^2 + 1}{t^2}} \right| \\ &= \left| \frac{t^2 - 1}{t^2 + 1} \right|\end{aligned}$$

$$\therefore \tan \angle POQ = \tan \theta \quad (\text{as required})$$

(iii) $S(c\sqrt{2}, c\sqrt{2})$

If $PS \perp OQ$, $\therefore m_{PS} = -1$

Hence $\left| \frac{\frac{c}{t} - c\sqrt{2}}{ct - c\sqrt{2}} \right| = -1$

Question 15 Solutions

(c)(iii)cont. $\frac{1}{t} - \sqrt{2} = -t + \sqrt{2}$

$$t + \frac{1}{t} = 2\sqrt{2}$$

$$\frac{t^2 + 1}{t} = 2\sqrt{2}$$

Now $\tan \theta = \tan \angle POQ$ from (ii),

$$\tan \theta = \frac{t^2 - 1}{t^2 + 1}$$

$$= \frac{\frac{t^2 - 1}{t}}{\frac{t^2 + 1}{t}}$$

$$= \frac{t - \frac{1}{t}}{t + \frac{1}{t}}$$

$$\therefore \tan^2 \theta = \left(\frac{t - \frac{1}{t}}{t + \frac{1}{t}} \right)^2$$

$$= \frac{t^2 - 2 + \frac{1}{t^2}}{\left(t + \frac{1}{t} \right)^2}$$

$$= \frac{\left(t^2 + 2 + \frac{1}{t^2} \right) - 2 - 2}{\left(t + \frac{1}{t} \right)^2}$$

$$= \frac{\left(t + \frac{1}{t} \right)^2 - 4}{\left(t + \frac{1}{t} \right)^2}$$

$$= \frac{(2\sqrt{2})^2 - 4}{(2\sqrt{2})^2}$$

$$= \frac{8 - 4}{8} = \frac{1}{2}$$

$$\therefore \tan \theta = \frac{1}{\sqrt{2}} \quad (\theta \geq 0) \quad (\text{as required})$$

Question 16 Solutions

$$\begin{aligned}
& \text{(a)(i)} \quad (\cos \theta + i \sin \theta)^5 \\
&= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 \\
&\quad + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\
&= \cos^5 \theta + 5i \cos^4 \theta \sin \theta + 10 \cos^3 \theta (i^2) \sin^2 \theta \\
&\quad + 10 \cos^2 \theta (i^3) \sin^3 \theta + 5 \cos \theta (i^4) \sin^4 \theta + (i^5) \sin^5 \theta \\
&= \cos^5 \theta + 5i \cos^4 \theta \sin \theta + 10 \cos^3 \theta (-1) \sin^2 \theta \\
&\quad + 10 \cos^2 \theta (-i) \sin^3 \theta + 5 \cos \theta (1) \sin^4 \theta + (i) \sin^5 \theta \\
&= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\
&\quad - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta
\end{aligned}$$

And by De Moivre's Theorem,

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

Equating the real terms:

$$\begin{aligned}
\cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\
&= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta (\sin^2 \theta)^2 \\
&= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\
&= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta \\
&\quad + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\
&= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta \\
&\quad + 5 \cos^5 \theta \\
\therefore \cos 5\theta &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad (\text{as required})
\end{aligned}$$

$$\begin{aligned}
& \text{(ii)} \quad \text{Let} \quad \cos 5\theta = 0 \\
& 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = 0 \quad \text{and let } x = \cos \theta
\end{aligned}$$

$$16x^5 - 20x^3 + 5x = 0$$

$$x(16x^4 - 20x^2 + 5) = 0$$

$$x = 0 \quad \text{or} \quad 16x^4 - 20x^2 + 5 = 0$$

The roots of x can be obtained from $\cos 5\theta = 0$;

$$5\theta = 2k\pi + \frac{\pi}{2} \quad \text{where } k = 0, 1, 2, 3, 4$$

$$5\theta = \frac{4k\pi + \pi}{2}$$

$$\therefore \theta = \frac{4k\pi + \pi}{10}$$

$$\text{When } k = 0, \quad \theta = \frac{\pi}{10}, \quad x_1 = \cos \frac{\pi}{10}$$

$$k = 1, \quad \theta = \frac{\pi}{2}, \quad x_2 = 0$$

$$k = 2, \quad \theta = \frac{9\pi}{10}, \quad x_3 = \cos \frac{9\pi}{10} = -\cos \frac{\pi}{10}$$

Question 16 Solutions

$$(a)(ii) \text{cont.} \quad k=3, \quad \theta = \frac{13\pi}{10}, \quad x_4 = \cos \frac{13\pi}{10} = -\cos \frac{3\pi}{10}$$
$$k=4, \quad \theta = \frac{17\pi}{10}, \quad x_5 = \cos \frac{17\pi}{10} = \cos \frac{3\pi}{10}$$

$\therefore 16x^4 - 20x^2 + 5 = 0$ has 4 non-zero roots are

$$\cos \frac{\pi}{10}, -\cos \frac{\pi}{10}, \cos \frac{3\pi}{10} \text{ and } -\cos \frac{3\pi}{10}.$$

$$(iii) \quad 16x^4 - 20x^2 + 5 = 0$$

$$x_1 x_3 x_4 x_5 = \frac{e}{a} \text{ where } e=5 \text{ and } a=16$$

$$x_1 x_3 x_4 x_5 = \frac{5}{16}$$

$$\cos \frac{\pi}{10} \left(-\cos \frac{\pi}{10} \right) \cos \frac{3\pi}{10} \left(-\cos \frac{3\pi}{10} \right) = \frac{5}{16}$$

$$\left(\cos \frac{\pi}{10} \cos \frac{3\pi}{10} \right)^2 = \frac{5}{16}$$

$$\therefore \cos \frac{\pi}{10} \cos \frac{3\pi}{10} = \frac{\sqrt{5}}{4} \quad \text{since } \cos \frac{\pi}{10} > 0$$

$$\text{and } \cos \frac{3\pi}{10} > 0$$

$$(iv) \quad \sin \frac{3\pi}{5} \sin \frac{6\pi}{5} = \cos \left(\frac{\pi}{2} - \frac{3\pi}{5} \right) \cos \left(\frac{\pi}{2} - \frac{6\pi}{5} \right)$$
$$= \cos \left(\frac{5\pi - 6\pi}{10} \right) \cos \left(\frac{5\pi - 12\pi}{10} \right)$$
$$= \cos \left(\frac{-\pi}{10} \right) \cos \left(\frac{7\pi}{10} \right)$$
$$= \cos \frac{\pi}{10} \left(-\cos \frac{3\pi}{10} \right)$$
$$= - \left(\cos \frac{\pi}{10} \cos \frac{3\pi}{10} \right)$$

$$\therefore \sin \frac{3\pi}{5} \sin \frac{6\pi}{5} = -\frac{\sqrt{5}}{4}$$

$$(b)(i) \quad \underline{\text{To prove:}} \quad \cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$$

$$\underline{\text{Proof:}} \quad \text{LHS} = \cos(\alpha + \beta) + \cos(\alpha - \beta)$$

$$= \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$+ \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= 2 \cos \alpha \cos \beta$$

$$\therefore \text{LHS} = \text{RHS} \quad (\text{QED})$$

Question 16 Solutions	Comment
(b)(ii) $\int \cos nx \cos mx \, dx$ $= \frac{1}{2} \int \cos(n+m)x + \cos(n-m)x \, dx$ $= \frac{1}{2} \left[\frac{\sin(n+m)x}{(n+m)} + \frac{\sin(n-m)x}{(n-m)} \right] + C$ $= \frac{\sin(n+m)x}{2(n+m)} + \frac{\sin(n-m)x}{2(n-m)} + C$ (iii) If $\alpha > \beta > 0$ $-2 \sin \alpha \sin \beta = \cos(\alpha + \beta) - \cos(\alpha - \beta)$ $\therefore 2 \sin \alpha \sin \beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$ $\therefore \sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$ $\sum_{r=1}^{r=9} \int_0^{\frac{\pi}{2}} \sin r x \sin x \, dx$ $= \int_0^{\frac{\pi}{2}} \sin x \sin x \, dx + \int_0^{\frac{\pi}{2}} \sin 2x \sin x \, dx + \int_0^{\frac{\pi}{2}} \sin 3x \sin x \, dx$ $+ \dots + \int_0^{\frac{\pi}{2}} \sin 8x \sin x \, dx + \int_0^{\frac{\pi}{2}} \sin 9x \sin x \, dx$ $= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos 0 - \cos 2x) \, dx + \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos x - \cos 3x) \, dx$ $+ \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos 2x - \cos 4x) \, dx + \dots + \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos 7x - \cos 9x) \, dx$ $+ \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos 8x - \cos 10x) \, dx$ $= \frac{1}{2} \left[\int_0^{\frac{\pi}{2}} \cos 0 - \cos 2x + \cos x - \cos 3x + \cos 2x - \cos 4x \right.$ $\left. + \dots + \cos 8x - \cos 9x - \cos 10x \, dx \right]$	

Question 16 Solutions	Comment
(b)(iii)cont. $= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos 0 + \cos x - \cos 9x - \cos 10x) \, dx$ $= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos x - \cos 9x - \cos 10x) \, dx$ $= \frac{1}{2} \left[x + \sin x - \frac{\sin 9x}{9} - \frac{\sin 10x}{10} \right]_0^{\frac{\pi}{2}}$ $= \frac{1}{2} \left[\left(\frac{\pi}{2} \right) + \sin \left(\frac{\pi}{2} \right) - \frac{\sin \left(\frac{9\pi}{2} \right)}{9} - \frac{\sin \left(\frac{10\pi}{2} \right)}{10} - 0 \right]$ $= \frac{1}{2} \left[\frac{\pi}{2} + 1 - \frac{(1)}{9} - \frac{\sin 5\pi}{10} \right]$ $= \frac{1}{2} \left[\frac{\pi}{2} + 1 - \frac{1}{9} + 0 \right]$ $= \frac{1}{2} \left[\frac{\pi}{2} + \frac{8}{9} \right]$ $= \frac{1}{2} \left(\frac{9\pi - 16}{18} \right)$ $= \frac{9\pi - 16}{36}$ ☺ THE END ☺	